Arduino Uno R3
5 V Pin Voltage Verification

Multimeter Inspection Procedure

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1. Purpose

This procedure describes how to verify that the **5 V output pin** of the Arduino Uno R3 supplies the correct regulated voltage, using a standard digital multimeter (DMM). The test is a non-destructive, in-circuit inspection and requires no firmware upload or code execution.

2. Background

When powered via USB, the Arduino Uno R3 receives 5 V directly from the host bus. When powered via the barrel jack, an onboard **NCP1117ST50T3G** linear voltage regulator steps the input voltage (7–12 V) down to a stable **5 V rail**. This rail is exposed on the **5V pin** (power header J1, pin 4) and serves as the primary supply for the ATmega328P microcontroller and all 5 V peripherals.

A voltage outside the specified tolerance indicates a fault in the USB supply, the onboard regulator, a solder defect, or an excessive load. This inspection must be performed before connecting any 5 V-sensitive peripheral.

3. Required Equipment

Item	Description	Specification / Notes
1	Digital Multimeter (DMM)	DC voltage mode, accuracy ±0.5% or better
2	Test leads	Standard banana-plug leads supplied with DMM
3	USB cable	USB Type-A to Type-B, connected to a powered USB host
4	Arduino Uno R3	Board under test – no shield or peripheral attached

NOTE: No peripheral, shield, or load must be connected to the 5V pin during this measurement. An external load will cause a voltage drop and produce a misleading reading.

4. Pin Location on the Board

Locate the **power header (J1)** on the Arduino Uno R3. It is the 8-pin single-row female header situated near the barrel jack connector. The **5V pin** is the **fourth pin from the end** closest to the barrel jack, labelled **"5V"** in the silkscreen.

Pin No.	Label	Function
J1-1	IOREF	I/O reference voltage (~5 V)
J1-2	RESET	Active-low reset

J1-3	3V3	3.3 V regulated output
J1-4	**5V**	**5 V supply ← TEST POINT**
J1-5	GND	Ground
J1-6	GND	Ground
J1-7	VIN	External supply input

5. Acceptance Criteria

The 5V pin output must fall within the following limits under no-load conditions:

Parameter	Min	Nominal	Max	Unit
5V Pin Output Voltage (USB powered, no load)	4.75	5.00	5.25	V DC
5V Pin Output Voltage (barrel jack powered)	4.75	5.00	5.25	V DC
Max Rated Output Current	—	—	500	mA

6. Test Procedure

Step 1 – Prepare the board

Ensure **no shield, peripheral, or external wiring** is attached to the Arduino. The board should be bare with only the USB cable connected.

Step 2 – Power the board

Connect the Arduino to a USB host (PC, laptop, or powered USB hub) using the USB Type-A to Type-B cable. Confirm the green **PWR LED** illuminates steadily. Do not use the barrel jack for this test.

Step 3 – Configure the DMM

Set the DMM selector to **DC Voltage (V—)** mode. If the meter is not auto-ranging, select the **10 V range**. Insert the red probe into the V/Ω terminal and the black probe into the COM terminal.

Step 4 – Connect the negative probe

Touch the **black (COM) probe** firmly to any **GND pin** on the board (e.g. J1-5 or J1-6, labelled GND). Maintain steady contact throughout the measurement.

Step 5 – Connect the positive probe

Touch the **red probe** firmly to the **5V pin (J1-4)**. Ensure the probe tip makes clean contact with the metal pin and does not bridge to an adjacent pin.

Step 6 – Read and record the voltage

Allow the DMM reading to stabilise (typically <2 seconds). Read and record the displayed voltage to two decimal places.

Step 7 – Evaluate the result

Compare the recorded voltage against the acceptance criteria in Section 5. A reading of **4.75 V – 5.25 V** constitutes a **PASS**. Any reading outside this range constitutes a **FAIL** — refer to Section 7.

Step 8 – Disconnect

Remove both probes and disconnect the USB cable. Document the result in the test record (Section 8).

> **WARNING:** Never connect the DMM probes across a live voltage source in resistance (Ω) or continuity mode. Always verify the DMM is set to DC Voltage before making contact with the board.

7. Fault Diagnosis

If the measured voltage falls outside 4.75 V – 5.25 V, perform the checks below in order before concluding that the board is defective.

Symptom	Likely Cause	Corrective Action
Voltage below 4.75 V (e.g. 4.2–4.6 V)	Excessive load, weak USB host, or degraded regulator	Disconnect all peripherals and re-measure. Try a different USB port or c
Voltage = 0 V	No power input, blown polyfuse, or open solder joint	Check USB cable and host port. Inspect polyfuse F1 for continuity. Boar
Voltage above 5.25 V (e.g. 5.4–5.8 V)	Regulator fault or incorrect barrel-jack supply	Verify barrel-jack input is within 7–12 V. If using USB and voltage is higl
Unstable / fluctuating	Poor probe contact, loose USB connector, or marginal supply	Clean probe tips with IPA and retry. Reseat USB cable. If still unstable, inspect

8. Test Record

Complete the following record upon finishing the inspection. Retain this record as part of the device history file.

Field	Entry
Board Serial / Asset No.	
Date of Inspection	
Inspector Name	
DMM Model & Serial No.	
DMM Last Calibration Date	
USB Supply Used	
Measured 5V Pin Voltage (V DC)	
Result	□ PASS (4.75–5.25 V) □ FAIL (out of range)

Remarks / Observations	
